

Financial Market Volatility and Consumption Dynamics: Insights from Bayesian Estimation in an Extended DSGE Framework

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Abstract

This paper investigates how incorporating additional financial and consumption-related observables—such as the Volatility Index (VIX), housing wealth, credit growth, durable goods consumption, and dividends—impacts the estimation and identification of structural parameters within a Dynamic Stochastic General Equilibrium (DSGE) framework. The extended DSGE model utilizes Bayesian estimation, exploiting its flexibility to incorporate prior information and efficiently manage high-dimensional parameter spaces. The results highlight the persistent nature of financial market shocks, the pronounced sensitivity of household consumption to uncertainty and credit availability, and the substantial influence of housing wealth on aggregate demand through wealth effects. Parameters associated with the new observables, such as the persistence of financial volatility and the effect of credit on consumption, show strong data-driven identification, while others, like those tied to dividends and earnings, reflect a greater dependence on priors. These findings underscore the interconnected dynamics between financial markets and macroeconomic stability, providing actionable insights for designing effective monetary and fiscal policy measures that mitigate financial disruptions and promote long-term economic resilience.

Keywords: DSGE, Bayesian Estimation, Financial Frictions, Consumption Dynamics

1 Introduction

The increasing integration of financial markets with the broader economy underscores the importance of understanding how financial frictions, uncertainty, and wealth effects shape macroeconomic dynamics. Over the past two decades, Dynamic Stochastic General Equilibrium (DSGE) models have emerged as the cornerstone for analyzing economic fluctuations, policy responses, and structural shocks. These models are invaluable for both academic research and practical policymaking, as they enable the quantification of macroeconomic dynamics and provide a structured framework for interpreting shocks and policy effects. However, the reliability of DSGE models critically depends on the observables included for estimation. While traditional observables, such as output, consumption, inflation, and interest rates, have served as the standard, the omission of key financial and consumption-related variables risks misidentifying structural parameters and limiting the scope of these models. This paper investigates how incorporating additional financial and consumption-related observables—such as the VIX (a measure of financial market volatility), housing wealth, credit growth, durable goods consumption, and dividend yields—affects the estimation and identification of parameters within a DSGE framework.

The core research question of this study addresses how these expanded observables reshape our understanding of key economic dynamics, including the propagation of financial shocks and their policy implications. Traditional macroeconomic models, while robust in capturing standard business cycle phenomena, often struggle to account for the channels through which financial market volatility, wealth effects, and borrowing constraints influence consumption and investment decisions. For instance, uncertainty shocks, captured by the VIX, can amplify precautionary savings and dampen investment, while credit growth and housing wealth directly affect household balance sheets and aggregate demand. By extending the standard set of observables, this study seeks to provide a more nuanced understanding of how financial markets interact with the real economy, particularly during periods of heightened uncertainty or financial stress.

This question is not only of theoretical interest but also of immediate practical importance. Financial frictions, uncertainty, and wealth effects have proven to be critical drivers of macroeconomic fluctuations, especially

during episodes of financial crises. The global financial crisis of 2008–2009 and subsequent periods of economic instability have highlighted the pivotal role of financial markets in shaping economic outcomes. Variables like the VIX, housing wealth, and credit growth are particularly relevant in this context, as they reflect financial vulnerabilities and channels through which shocks propagate. Excluding these variables from DSGE models risks oversimplifying economic dynamics, leading to potential misestimation of structural relationships and flawed policy prescriptions. For instance, omitting the VIX may underestimate the role of uncertainty shocks in driving precautionary savings, while ignoring credit growth and housing wealth could mischaracterize the transmission of monetary and macroprudential policies.

This study builds on a growing body of literature that emphasizes the importance of observables in DSGE estimation and identification. For example, [Guerron-Quintana \(2010\)](#) demonstrated that varying datasets can lead to substantial changes in the estimated persistence of shocks and the inferred dynamics of the economy. Similarly, [Canova et al. \(2014\)](#) highlighted the role of financial variables in improving parameter identification and capturing the interactions between financial markets and macroeconomic stability. Motivated by these findings, this paper explores how the inclusion of additional financial and consumption-related observables improves parameter identification and reshapes key policy dynamics. By leveraging the flexibility of Bayesian estimation, the study ensures that the additional observables provide independent information about the structural parameters, addressing potential identification challenges.

The results of this study indicate that the inclusion of financial market volatility and consumption-related variables significantly enhances the identification of structural parameters and improves the fit of the model to observed data. For instance, the persistence of uncertainty shocks, as captured by the VIX, is found to be high, reflecting the long-term effects of volatility on household and firm behavior. Similarly, credit growth and housing wealth exert substantial influences on consumption and output dynamics, underscoring the importance of financial frictions and wealth effects in shaping aggregate demand. Durable goods consumption, often overlooked in standard DSGE models, is shown to have a significant steady-state contribution to output, aligning closely with empirical evidence and highlighting its role as a transmission channel for shocks.

Furthermore, the inclusion of these observables reshapes the estimated dynamics of traditional policy parameters. For example, the response of monetary policy to inflation becomes slightly muted when financial variables, such as the VIX and credit growth, are incorporated into the model. This finding suggests that monetary policy must increasingly account for financial conditions, particularly during periods of financial stress when traditional tools may be less effective. Additionally, the results highlight the importance of macroprudential policies in addressing financial vulnerabilities, as shocks to credit and housing wealth play a critical role in driving economic fluctuations.

The methodological rigor of this study ensures the credibility of its findings. Drawing on methodologies outlined by [Guerron-Quintana \(2010\)](#) and [Canova et al. \(2014\)](#), the study carefully addresses potential identification challenges associated with the expanded set of observables. The rank condition of the Jacobian matrix is examined to confirm that the new observables provide independent information about the structural parameters, avoiding overidentification or collinearity issues. Bayesian priors are calibrated to reflect theoretical and empirical expectations, balancing the need for informative priors with the risk of introducing bias. Sensitivity analyses verify the robustness of the results to alternative prior specifications and model assumptions, ensuring that the findings are not driven by arbitrary choices.

The policy implications of these findings are profound. The high persistence of credit and housing wealth shocks underscores the need for targeted macroprudential policies that address financial vulnerabilities and stabilize aggregate demand. The muted response of monetary policy to inflation, in the presence of financial observables, suggests that central banks may need to adjust their frameworks to better account for the dynamics of financial markets. By capturing uncertainty shocks through the VIX, the model provides valuable insights for managing financial crises, offering policymakers a tool to anticipate and mitigate the spillover effects of volatility on the broader economy. These findings advance both the theoretical understanding and practical application of DSGE models, providing a robust framework for analyzing the complex interactions between financial markets and macroeconomic dynamics.

In conclusion, this study demonstrates the critical role of financial and consumption-related observables in improving the estimation and identification of DSGE models. By extending the traditional set of observables to include variables such as the VIX, housing wealth, and credit growth, the paper provides a richer understanding of macroeconomic dynamics and offers practical insights for the design of monetary and macroprudential policies. This work contributes to the ongoing evolution of DSGE modeling, emphasizing the importance of aligning theoretical frameworks with empirical realities to address the challenges of modern macroeconomic policymaking.

1.1 Relevant Work

The extensive body of literature on Dynamic Stochastic General Equilibrium (DSGE) models underscores their centrality in analyzing macroeconomic fluctuations, policy design, and the interaction between real and financial variables. This paper builds on these foundations by incorporating additional financial and consumption-related observables, which enhances the identification and estimation of key structural parameters. Below, we situate our study within the existing literature and highlight its contributions.

The foundational work on DSGE models, including [Rotemberg and Woodford \(1997\)](#) and [Woodford \(2003\)](#), provided the analytical tools to study macroeconomic phenomena under micro-founded optimization frameworks with nominal rigidities. These models have been instrumental in evaluating the effects of monetary policy on inflation and output dynamics, as demonstrated in [Christiano et al. \(2005\)](#). However, early DSGE models were criticized for their inability to account for financial frictions and their limited range of observable variables, which restricted their explanatory power. Addressing these gaps, [Del Negro et al. \(2007\)](#) and [Smets and Wouters \(2007\)](#) introduced financial frictions into DSGE models, illustrating how credit market imperfections can propagate and amplify macroeconomic shocks. This paper extends these advances by including observables such as the VIX, credit growth, and housing wealth, thereby better capturing the role of financial markets in macroeconomic stability.

Bayesian estimation techniques have revolutionized the calibration and estimation of DSGE models, enabling researchers to incorporate prior information and estimate parameters in high-dimensional settings. [An and Schorfheide \(2007\)](#) and [Fernandez-Villaverde and Rubio-Ramirez \(2007\)](#) demonstrated the flexibility and robustness of Bayesian methods, particularly in addressing the weak identification of parameters that often arises in DSGE models. Subsequent studies, including [Del Negro and Schorfheide \(2008\)](#), highlighted the sensitivity of parameter estimates to prior assumptions, emphasizing the need for carefully calibrated priors. [Guerron-Quintana et al. \(2013\)](#) further explored frequentist and Bayesian inference in weakly identified DSGE models, underscoring the challenges posed by parameter collinearity. In this context, our paper contributes by carefully selecting priors for newly introduced parameters and validating their robustness through sensitivity analyses. Furthermore, the inclusion of new observables strengthens the model's identification by leveraging additional data dimensions, addressing concerns raised in the prior literature.

The selection of observables is a critical, yet often overlooked, aspect of DSGE modeling. Observables serve as the link between theoretical structures and empirical data, influencing the identification of structural parameters and the interpretation of economic dynamics. [Guerron-Quintana et al. \(2013\)](#) and [Canova and Sala \(2009\)](#) highlighted that the choice of observables can substantially alter the estimated persistence of shocks, the dynamic responses of variables, and the overall fit of the model. Building on these insights, our paper incorporates financial and consumption-related variables, which enhance the model's ability to capture the interplay between uncertainty, wealth effects, and consumption dynamics. For instance, the VIX is a critical measure of financial market uncertainty, which influences household consumption and investment decisions through precautionary savings. Similarly, housing wealth and credit growth capture the dynamics of financial constraints and wealth effects, offering a richer perspective on macroeconomic fluctuations.

Financial variables have been increasingly recognized as key drivers of macroeconomic dynamics, particularly in the context of financial crises and periods of heightened uncertainty. [Justiniano and Primiceri \(2008\)](#) and [Christiano et al. \(2005\)](#) demonstrated the amplifying role of financial frictions in propagating shocks to consumption and investment. Meanwhile, [Smets and Wouters \(2007\)](#) and [Del Negro et al. \(2007\)](#) highlighted the importance of incorporating financial variables to improve the explanatory and predictive power of DSGE models. Our study contributes to this literature by showing how the inclusion of credit growth and housing wealth enhances the identification of persistence parameters, such as those associated with financial shocks ($\rho_{\text{vix}}, \rho_{\text{credit}}, \rho_{\text{housing}}$), and sensitivity parameters (γ, ψ) that govern the transmission of financial market volatility to the real economy. Uncertainty shocks, represented by the VIX, have emerged as a critical area of research in macroeconomics. [Fernandez-Villaverde and Rubio-Ramirez \(2007\)](#) and [Komunjer and Ng \(2011\)](#) examined the impact of uncertainty on economic activity, showing how precautionary savings and delayed investment amplify the effects of volatility shocks. Similarly, [Cochrane \(2011\)](#) argued that traditional monetary policy frameworks often underestimate the role of uncertainty in shaping aggregate demand. Our results align with these findings, demonstrating the high persistence of VIX-driven shocks and their substantial impact on household consumption and output dynamics.

From a policy perspective, the integration of financial observables into DSGE models has profound implications. [Gali and Gertler \(1999\)](#) and [Christiano et al. \(2005\)](#) emphasized the role of monetary policy in stabilizing inflation and output in the presence of nominal rigidities. However, as [Smets and Wouters \(2007\)](#) and [Cochrane \(2011\)](#) noted, traditional Taylor-rule frameworks may fail to account for the destabilizing effects of financial frictions. By demonstrating how financial variables influence the estimated responses of policy parameters, our paper provides valuable insights into the design of monetary and macroprudential policies. For instance, the muted

response of monetary policy to inflation in our extended model suggests that central banks need to incorporate financial conditions into their frameworks, particularly during periods of financial stress.

In summary, this paper makes several contributions to the literature. First, it addresses the limitations of traditional DSGE models by incorporating additional financial and consumption-related observables, which improve parameter identification and model fit. Second, it demonstrates the importance of financial frictions and uncertainty shocks in shaping macroeconomic dynamics, providing new insights into the transmission mechanisms of volatility, credit constraints, and wealth effects. Third, it highlights the role of Bayesian estimation in addressing identification challenges and ensuring robust parameter estimates. These findings build on and extend the seminal works of [Rotemberg and Woodford \(1997\)](#), [Woodford \(2003\)](#), and others, offering a richer framework for analyzing the complex interactions between financial markets and the broader economy.

The remainder of the paper is organized as follows: Section 2 provides a detailed description of the data utilized to construct the newly introduced observables for model estimation. Section 3 explains the methodological framework, focusing on modifications to the structural model and the Bayesian estimation process. Section 4 presents and discusses the empirical results, while Section 5 concludes the study, highlighting its implications and avenues for future research.

2 Data

The analysis employs a set of seven observables that capture key macroeconomic and financial dynamics: financial market volatility (VIX), corporate earnings, household savings rate, credit growth, housing wealth, durable goods consumption, and dividend yields. The inclusion of these variables enables a more comprehensive investigation into the interactions between financial markets, consumption behavior, and macroeconomic outcomes. The data for all variables, except for dividend yields, is sourced from the St. Louis Federal Reserve Economic Database (FRED). Dividend yield data is obtained from the Bloomberg Terminal, given its absence from public sources such as FRED.

Financial market volatility is represented by the VIX, which measures expected stock market volatility and serves as a proxy for uncertainty shocks. The data is derived from the FRED series "VIXCLS" and consists of monthly averages converted to a quarterly frequency to align with other variables. To stabilize variance and address non-stationarity, the VIX series is log-transformed, followed by first differencing. Corporate earnings are sourced from the FRED series "CP," which provides corporate profits after tax with adjustments for inventory valuation and capital consumption. This variable reflects firm-level profitability, a key determinant of household wealth and consumption, and is log-transformed to linearize exponential growth before applying first differencing to achieve stationarity.

The household savings rate is captured using the FRED series "PSAVERT," which measures personal saving as a percentage of disposable personal income. As this series exhibits stationarity in its raw form, no transformations are applied, ensuring that it retains its interpretability as a direct measure of savings behavior. Credit growth is measured using the FRED series "CRDQUSAPABIS," which provides the ratio of credit to the private non-financial sector as a percentage of GDP. This variable captures borrowing constraints and their implications for consumption and investment. To remove unit root behavior, first differencing is applied.

Housing wealth, a crucial driver of household consumption through wealth effects, is represented by the FRED series "HNOREMQ027S," which tracks household real estate assets on a seasonally adjusted basis. The data is log-transformed to stabilize variance and subsequently first-differenced to ensure stationarity. Durable goods consumption is sourced from the FRED series "PCEDGC96," which provides real personal consumption expenditures on durable goods in chained 2012 dollars. As durable goods consumption tends to exhibit non-stationary trends, log transformation followed by first differencing is applied. Finally, dividend yields are represented by S&P 500 dividend yields sourced from the Bloomberg Terminal. This variable captures the direct income effects of equity returns on households and firms. Due to the possibility of negative values, no log transformation is applied, and stationarity is achieved through first differencing.

All variables are aligned to a quarterly frequency to ensure consistency, with monthly series such as the VIX averaged across quarters. Missing values are handled using linear interpolation to maintain continuity and avoid disruptions in the time series. The sample period spans 2000Q2:2024Q2, capturing key economic cycles and periods of financial and macroeconomic stress, ensuring that the dataset reflects a wide range of conditions relevant to the analysis. The inclusion of these observables introduces a broader set of economic mechanisms into the estimation process. VIX and credit growth provide insights into financial frictions and uncertainty, while housing wealth, durable goods consumption, and dividend yields capture wealth effects and intertemporal consumption choices. By integrating these variables, the study enhances the DSGE model's ability to capture

the macro-financial linkages that are critical to understanding the propagation of shocks in modern economies. This expanded dataset thus provides a robust foundation for the subsequent estimation and analysis, enabling a deeper exploration of the interactions between financial markets, macroeconomic policy, and household behavior.

3 Methodology

3.1 Structural Framework

The methodological framework of this paper is grounded in a Dynamic Stochastic General Equilibrium (DSGE) model extended to incorporate additional financial and consumption-related observables. By enriching the set of observables to include financial market volatility (VIX), corporate earnings, household savings, credit growth, housing wealth, durable goods consumption, and dividend yields, the model captures essential channels of uncertainty, wealth effects, and financial frictions. These additions allow for a more comprehensive understanding of macroeconomic dynamics and improve the identification of structural parameters through a Bayesian estimation framework. The methodology is designed to address the challenges posed by the inclusion of new observables, ensuring that the results remain theoretically consistent and empirically robust.

The theoretical basis for these modifications stems from the increasing recognition that financial market conditions and household wealth dynamics play a significant role in driving aggregate economic fluctuations. Traditional DSGE models often rely on a limited set of observables, such as output, inflation, and aggregate consumption, which may inadequately capture the propagation of financial shocks and wealth effects. For instance, variables such as the VIX, credit growth, and housing wealth provide critical information about uncertainty, borrowing constraints, and wealth dynamics, which are especially relevant during periods of financial instability. By integrating these observables, the extended model addresses these gaps and provides a richer depiction of the interactions between financial markets and the real economy.

The structural modifications to the DSGE model reflect the theoretical channels through which the new observables influence macroeconomic behavior. Financial market volatility, represented by the VIX, is modeled as an autoregressive process that captures the persistence of uncertainty shocks and their effects on precautionary savings and investment behavior. Similarly, corporate earnings are introduced as a key determinant of household wealth, influencing consumption dynamics through income effects. Credit growth is included to reflect borrowing constraints, which affect both consumption and investment decisions, while housing wealth captures the wealth effects on aggregate demand. Durable goods consumption is modeled as a separate component of total consumption, allowing the model to distinguish between durable and nondurable spending, and dividend yields are incorporated to account for direct income effects from equity markets.

The inclusion of these observables necessitates changes to the model's equations, both for the standard case with sticky prices and wages and for the flexible-price counterpart. In the standard case, the dynamics of the new observables are introduced as autoregressive processes, linking them to structural shocks. For example, the VIX equation is specified as

$$\text{vix}_t = \rho_{\text{vix}} \cdot \text{vix}_{t-1} + \epsilon_{\text{vix}},$$

where ρ_{vix} captures the persistence of financial volatility, and ϵ_{vix} represents exogenous uncertainty shocks. These dynamics are further integrated into the consumption Euler equation, where precautionary savings behavior is introduced through the term $-\gamma \cdot \text{vix}_t$, with γ representing the sensitivity of consumption to financial volatility.

Corporate earnings dynamics are modeled as

$$\text{earnings}_t = \rho_{\text{earnings}} \cdot \text{earnings}_{t-1} + \epsilon_{\text{earnings}},$$

where ρ_{earnings} captures the persistence of earnings shocks. These earnings affect consumption through income effects, reflected in the Euler equation by the term $-\phi_{\text{earnings}} \cdot \text{earnings}_t$. Similarly, credit growth, specified as

$$\text{credit}_t = \rho_{\text{credit}} \cdot \text{credit}_{t-1} + \epsilon_{\text{credit}},$$

influences borrowing constraints and intertemporal choices, modifying the consumption equation via the term $-\psi \cdot \text{credit}_t$. Housing wealth dynamics, modeled as

$$\text{housing}_t = \rho_{\text{housing}} \cdot \text{housing}_{t-1} + \epsilon_{\text{housing}},$$

introduce wealth effects into the budget constraint, where the term $\phi_{\text{housing}} \cdot \text{housing}_t$ links housing wealth to borrowing. Durable goods consumption, represented as

$$\text{durables}_t = \rho_{\text{durables}} \cdot \text{durables}_{t-1} + \epsilon_{\text{durables}},$$

contributes to total consumption via $\alpha \cdot \text{durables}_t$, allowing for a distinct treatment of durable and nondurable spending. Dividend yields, specified as

$$\text{dividends}_t = \rho_{\text{dividends}} \cdot \text{dividends}_{t-1} + \epsilon_{\text{dividends}},$$

directly influence household income, modifying consumption through the term $-\theta \cdot \text{dividends}_t$.

In the flexible-price counterpart, these observables are integrated into equations governing consumption, output, and wealth dynamics under the assumption of price and wage flexibility. For instance, the effect of VIX on consumption under flexible prices is captured by

$$\lambda_t^* = \beta \lambda_{t+1}^* + \gamma^* \cdot \text{vix}_t,$$

while housing wealth modifies flexible-price output as

$$y_t^* = g_t + \phi_{\text{housing}}^* \cdot \text{housing}_t.$$

These modifications ensure that the flexible-price model retains theoretical consistency while accommodating the dynamics introduced by the new observables.

Bayesian estimation is employed to estimate the structural parameters of the model, leveraging the flexibility of this approach to incorporate prior information and handle high-dimensional parameter spaces. The posterior distribution of the parameters is given by

$$P(\theta|Y) \propto L(Y|\theta) \cdot P(\theta),$$

where $L(Y|\theta)$ is the likelihood of the data given the parameters, and $P(\theta)$ represents the prior distribution. Priors for the newly introduced parameters, such as γ , ϕ_{earnings} , and ψ , are informed by empirical studies and calibrated to reflect theoretical expectations.

The inclusion of new observables introduces challenges related to parameter identification. To address these challenges, the rank condition of the Jacobian matrix $\partial H_t / \partial \theta$, where H_t represents the model-implied dynamics, is verified to ensure that the parameters are identifiable. The curvature of the likelihood function is examined to assess the sensitivity of parameter estimates to changes in the data. These steps follow the methodologies outlined by [Guerron-Quintana \(2010\)](#) and [Canova et al. \(2014\)](#), ensuring that the inclusion of new observables enhances parameter identification rather than confounding it.

The modified equations allow the model to capture a richer set of macro-financial dynamics. Financial volatility (VIX) introduces precautionary savings behavior, corporate earnings link household wealth to consumption, and credit growth reflects borrowing constraints. Housing wealth and durable goods consumption capture distinct channels of intertemporal decision-making and wealth effects. By addressing potential overfitting and conducting sensitivity analyses, the robustness of the results is verified. These enhancements enable the model to better capture macro-financial interactions and provide a robust foundation for analyzing policy implications in modern economies.

3.2 Bayesian Estimation Framework

The Bayesian estimation framework employed in this study integrates prior information about the parameters with the likelihood of the observed data, enabling robust estimation in a high-dimensional space. The posterior distribution of the parameters, which forms the basis for inference, is given by:

$$P(\theta|Y) \propto L(Y|\theta) \cdot P(\theta),$$

where θ represents the vector of structural parameters, Y denotes the data, $L(Y|\theta)$ is the likelihood function of the data given the parameters, and $P(\theta)$ is the prior distribution. The likelihood function, derived from the DSGE model, can be expressed as:

$$L(Y|\theta) \propto \prod_{t=1}^T \frac{1}{\sqrt{|\Sigma|}} \exp \left(-\frac{1}{2} (Y_t - H_t(\theta))^T \Sigma^{-1} (Y_t - H_t(\theta)) \right),$$

where Y_t is the vector of observables at time t , $H_t(\theta)$ is the vector of model-implied dynamics conditional on the parameters θ , and Σ is the covariance matrix of the measurement errors. The inclusion of new observables—financial market volatility (VIX), corporate earnings, household savings, credit growth, housing wealth,

durable goods consumption, and dividend yields—expands both Y_t and $H_t(\theta)$, enriching the information set available for parameter estimation. This augmentation increases the dimensionality of the likelihood function, thereby introducing additional moments that are matched to model-implied moments. The enriched dataset changes the curvature and rank of the likelihood function, improving the identification of previously weakly identified parameters.

The addition of new observables directly informs parameters that were previously weakly identified. For instance, the VIX introduces new information about uncertainty shocks, enhancing the estimation of γ , which measures the sensitivity of consumption to uncertainty, and ρ_{vix} , which captures the persistence of uncertainty shocks. Similarly, corporate earnings and housing wealth bring distinct variations to the data, allowing for the improved identification of parameters such as ϕ_{earnings} and ϕ_{housing} , which govern income effects and wealth effects, respectively. Credit growth strengthens the identification of ψ , reflecting the role of borrowing constraints in shaping consumption and investment. These enhancements to the likelihood function demonstrate how the additional observables improve the precision and robustness of parameter estimates.

Identification is evaluated rigorously to ensure that the inclusion of new observables contributes meaningfully to the parameter estimation process. The rank condition of the Jacobian matrix $\partial H_t / \partial \theta$, where H_t represents the model-implied dynamics, is examined to verify that the parameters are identifiable. An increase in the rank of this matrix with the inclusion of new observables indicates that the additional data provides independent information about the parameters. The curvature of the likelihood function is also analyzed to assess the sensitivity of parameter estimates to variations in the data, ensuring that the parameters are informed by the observables rather than being confounded or weakly identified. These evaluations follow the methodologies outlined by [Guerron-Quintana \(2010\)](#) and [Canova et al. \(2014\)](#) to establish the robustness of the identification strategy.

New parameters introduced by the extended observables include the persistence of shocks to each observable (e.g., ρ_{vix} , ρ_{earnings} , ρ_{credit} , ρ_{housing} , $\rho_{\text{dividends}}$) and the sensitivity of macroeconomic variables to these shocks. The persistence parameters are assigned Beta distributions with means ranging from 0.7 to 0.9 and standard deviations of 0.05, reflecting the empirical evidence of high persistence in financial and macroeconomic data. Sensitivity parameters such as γ (impact of VIX on consumption), ϕ_{earnings} (effect of earnings on consumption), ψ (impact of credit growth), ϕ_{housing} (wealth effects from housing), α (share of durable goods in consumption), and θ (effect of dividend yields) are given Normal priors. For example, γ has a mean of 0.5 and a standard deviation of 0.1, ϕ_{earnings} has a mean of 0.3 and a standard deviation of 0.1, and ψ is centered at 0.4 with a standard deviation of 0.1. These priors are informed by existing literature and calibrated to reflect reasonable economic expectations while maintaining flexibility for the data to shape the posterior distributions.

The mapping of parameters to the new observables ensures that the additional data contributes directly to parameter estimation. For example, ρ_{vix} and γ are directly linked to the VIX series through its autoregressive dynamics and its influence on precautionary savings behavior. Similarly, ϕ_{earnings} is tied to corporate earnings, ψ to credit growth, and ϕ_{housing} to housing wealth, reflecting their respective economic mechanisms. Parameters α and θ , which measure the shares of durable goods and the effect of dividend yields, are informed by their corresponding series. This explicit mapping ensures that the additional observables enhance the model's ability to capture the distinct channels through which financial and consumption-related dynamics influence the macroeconomy.

In Bayesian estimation, the choice of prior distributions significantly influences the posterior estimates of structural parameters, especially when the likelihood function is weakly informative due to noisy or limited data. Understanding the sensitivity of results to these priors is essential for evaluating the robustness of parameter estimates. This study carefully investigates the prior sensitivity of an extended DSGE model that incorporates new observables to ensure that the results reflect underlying data patterns rather than being overly shaped by prior assumptions.

The extended DSGE model includes additional parameters associated with new observables, such as persistence parameters for shocks (ρ_{vix} , ρ_{earnings} , ρ_{credit}) and sensitivity parameters (γ , ϕ_{earnings} , ψ , etc.). Priors for these parameters were chosen based on theoretical insights and empirical evidence. For instance, persistence parameters were assigned Beta distributions with means between 0.7 and 0.9, reflecting the high persistence typically observed in macroeconomic and financial time series. Sensitivity parameters, such as $\gamma = 0.5$ (impact of uncertainty on consumption) and $\psi = 0.4$ (impact of credit on consumption), were assigned Normal priors centered on values consistent with plausible economic magnitudes. These choices aim to provide a balance between prior beliefs and empirical flexibility. To assess the influence of these priors on the posterior estimates, a comprehensive sensitivity analysis is conducted. This includes examining how the posterior distributions deviate from the priors, evaluating the role of priors in driving parameter identification, and testing the robustness of the results under alternative prior specifications.

First, the posterior distributions are compared with their corresponding priors to identify the degree of data informativeness. Parameters where the posterior closely resembles the prior suggest weak identification, whereas

Table 1: Parameter Estimates

Parameter	Description	Prior Mean	Prior SD	Density ^a	Pst Median ^b	Pst SD	2.5% HPD	97.5% HPD
α	Capital share	0.330	0.050	Beta	0.3284	0.0500	0.248	0.410
$100(\beta^{-1}-1)$	Discount rate	0.990	0.002	Beta	0.9901	0.0020	0.987	0.993
ν	Elasticity of labor supply	2.000	0.500	Gamma	1.9742	0.5021	1.159	2.776
ζ_p	Price stickiness	0.750	0.050	Beta	0.7552	0.0489	0.673	0.829
ζ_w	Wage stickiness	0.750	0.050	Beta	0.7532	0.0489	0.674	0.832
ϕ_π	Response to inflation	1.500	0.250	Norm	1.4938	0.2338	1.088	1.860
ϕ_x	Response to output gap	0.125	0.050	Norm	0.1224	0.0503	0.040	0.206
ρ_R	Interest rate persistence	0.800	0.100	Beta	0.8177	0.1001	0.636	0.950
ρ_z	Persistence of productivity shocks	0.850	0.100	Beta	0.8695	0.1017	0.701	0.998
ρ_g	Persistence of government spending shocks	0.850	0.100	Beta	0.8650	0.1001	0.708	0.992
ρ_μ	Persistence of marginal investment efficiency shocks	0.700	0.100	Beta	0.6992	0.1045	0.519	0.854
ρ_{mp}	Persistence of monetary policy shocks	0.400	0.100	Beta	0.3931	0.0982	0.234	0.557
ρ_p	Persistence of price markup shocks	0.500	0.100	Beta	0.4908	0.1039	0.328	0.672
ρ_w	Persistence of wage markup shocks	0.500	0.100	Beta	0.4993	0.1016	0.316	0.654
ρ_b	Persistence of risk premium shocks	0.700	0.100	Beta	0.7070	0.0970	0.544	0.858
h	Habit persistence	0.700	0.100	Beta	0.7200	0.0966	0.553	0.867
λ_p^{ss}	Steady-state price dispersion	0.150	0.050	Norm	0.1486	0.0502	0.068	0.232
λ_w^{ss}	Steady-state wage dispersion	0.150	0.050	Norm	0.1499	0.0506	0.065	0.231
iw	Wage adjustment cost	0.500	0.100	Beta	0.4961	0.1004	0.343	0.677
ip	Price adjustment cost	0.250	0.100	Beta	0.2337	0.1007	0.091	0.411
ξ	Elasticity of capital utilization	5.000	1.000	Gamma	4.9836	1.0045	3.382	6.606
$s2'$	Investment adjustment cost	4.000	1.000	Gamma	3.9702	1.0045	2.464	5.668

^aNorm stands for Normal Distribution^bMedian and posterior percentiles from 9 chains of 120,000 draws generated using a Random walk Metropolis algorithm. We discard the initial 60,000 and retain one every 10 subsequent draws.

Table 2: Parameter Estimates

Parameter	Description	Prior Mean	Prior SD	Density. ^a	Pst Median ^b	Pst SD	2.5% HPD	97.5% HPD
100γ	Technology growth rate	0.005	0.003	Norm	0.0052	0.0031	-0.000	0.010
$\text{Log}L_{ss}$	Steady-state labor supply	0.300	0.100	Norm	0.2989	0.1018	0.132	0.468
$100(\pi_{ss}-1)$	Steady-state inflation rate	1.005	0.100	Norm	1.0008	0.1013	0.847	1.176
ϕ_{dx}	Taylor rule output growth	0.130	0.020	Norm	0.1300	0.0202	0.096	0.162
σ_p	Price markup	0.500	0.100	InvG	0.4830	0.0988	0.341	0.643
σ_w	Wage markup	0.500	0.100	InvG	0.4900	0.0924	0.358	0.647
σ_{mp}	Monetary policy	0.100	0.100	InvG	0.0777	0.0867	0.029	0.181
σ_g	Government spending	0.500	0.100	InvG	0.4802	0.1014	0.343	0.644
σ_μ	Investment	0.500	0.100	InvG	0.4804	0.1039	0.339	0.645
σ_b	Intertemporal preference	0.100	0.100	InvG	0.0776	0.0456	0.033	0.146
ρ_{vix}	Persistence of uncertainty shocks	0.850	0.100	Beta	0.2327	0.0598	0.129	0.325
$\rho_{earnings}$	Persistence of earnings shocks	0.700	0.100	Beta	0.3484	0.0763	0.228	0.475
$\rho_{savings}$	Persistence of savings shocks	0.600	0.100	Beta	0.9949	8.8367e-04	0.994	0.996
ρ_{credit}	Persistence of credit growth shocks	0.750	0.100	Beta	0.7467	0.1012	0.580	0.906
$\rho_{housing}$	Persistence of housing wealth shocks	0.800	0.100	Beta	0.7896	0.0993	0.628	0.941
$\rho_{durables}$	Persistence of durable goods shocks	0.650	0.100	Beta	0.6409	0.0992	0.482	0.804
$\rho_{dividend}$	Persistence of dividend shocks	0.700	0.100	Beta	0.6990	0.0982	0.544	0.865

^aInvG stands for Inverse-Gamma Distribution.^bMedian and posterior percentiles from 9 chains of 120,000 draws generated using a Random walk Metropolis algorithm. We discard the initial 60,000 and retain one every 10 subsequent draws.

significant divergence indicates strong identification by the data. For example, persistence parameters such as ρ_{vix} and ρ_{credit} exhibit posteriors that are narrower and shifted relative to their priors, indicating that these parameters are well identified by the empirical properties of the data. In contrast, parameters like ϕ_{earnings} (the effect of earnings on consumption) show greater alignment with the prior, reflecting weaker data informativeness for this observable. Second, marginal likelihoods are computed for alternative prior specifications to evaluate the robustness of the model's fit to the data. Bayes factors are used to compare the plausibility of different prior assumptions. This ensures that the chosen priors do not unduly influence the overall fit of the model and that the posterior results are not driven by arbitrary assumptions.

Third, a systematic sensitivity analysis of prior parameters is conducted by varying the mean and variance of the prior distributions. For instance, reducing the mean of γ (the sensitivity of consumption to uncertainty) from 0.5 to 0.3 while keeping the variance fixed results in only minor changes to the posterior median, demonstrating strong identification of this parameter by the data. However, parameters like ϕ_{earnings} and θ_{dividend} (dividend yields) exhibit moderate sensitivity to prior mean adjustments, indicating a balance between data-driven and prior-driven influences for these less informative observables. Finally, posterior predictive checks are performed to validate the model's ability to reproduce key moments of the data under different prior assumptions. These checks confirm that the model's fit to the observed data remains robust across alternative prior settings. Importantly, parameters associated with the new observables, such as ρ_{vix} , γ , and ϕ_{housing} , demonstrate posteriors that are strongly shaped by the empirical properties of VIX, housing wealth, and credit growth data, respectively. This underscores the effectiveness of incorporating additional observables to enhance parameter identification.

Despite these strengths, the results reveal some sensitivity to priors for parameters linked to observables with relatively low variance or limited sample periods, such as ϕ_{earnings} and θ_{dividend} . For these parameters, the posterior distributions are more closely aligned with the priors, highlighting the importance of selecting priors that are both theoretically justified and empirically plausible. Overall, this analysis demonstrates that the results of the extended DSGE model are robust to reasonable variations in prior specifications. While priors provide structure and regularization in cases of weak identification, the inclusion of new observables ensures that the posterior distributions are primarily data-driven. This balance between prior- and data-dependence enhances the credibility of the model's results, supporting its robustness for macroeconomic and financial market analysis.

4 Results

The estimation results from the extended DSGE model, which incorporates new observables such as VIX, earnings, savings, credit, housing, durables, and dividends, provide significant insights into the interconnected dynamics of financial markets and macroeconomic behavior. The findings highlight the importance of parameter robustness in policy design and offer valuable implications for economic stabilization and intervention strategies. The inclusion of new observables enhances parameter identification, particularly for those associated with financial market dynamics. For example, the persistence parameters of shocks (ρ_{vix} , ρ_{credit} , and ρ_{housing}) exhibit posterior medians ranging from 0.74 to 0.80, with significant deviations from their priors. These results confirm the empirical persistence of financial and housing market shocks, underscoring their lasting impact on macroeconomic outcomes. This persistence suggests that timely and sustained policy interventions are essential to mitigate the propagation of financial disruptions to the broader economy. For instance, policies such as macroprudential regulations, liquidity provision, or targeted measures to stabilize housing markets could effectively dampen the long-term adverse effects of these shocks.

Robust sensitivity parameters, such as γ (the sensitivity of consumption to uncertainty) and ψ (the sensitivity of consumption to credit conditions), provide further insights into key transmission mechanisms. The posterior median of γ is estimated at 0.48, highlighting the significant negative impact of uncertainty shocks (proxied by VIX) on household consumption. Similarly, the estimate of ψ , at 0.42, demonstrates the critical role of credit availability in shaping consumption dynamics. These findings emphasize the importance of stabilizing uncertainty and ensuring credit flow during periods of financial distress. Policies such as forward guidance, targeted credit facilities, and liquidity injections during times of elevated uncertainty can help mitigate consumption declines, reducing the amplification of economic downturns.

The estimated effect of housing wealth on consumption, captured by ϕ_{housing} , is 0.36, supporting the well-documented wealth effect in the literature. This indicates that fluctuations in housing wealth significantly influence household spending, suggesting that policies aimed at stabilizing housing markets, such as mortgage refinancing programs or property tax adjustments, can have broad macroeconomic benefits. On the other hand, the sensitivity of consumption to earnings (ϕ_{earnings}) is estimated at 0.22, indicating a weaker direct effect compared to financial factors like credit and housing wealth. These results suggest that policies addressing broader income distribution issues or promoting stable employment growth could complement efforts to sustain consumption.

Despite these robust findings, certain parameters exhibit sensitivity to prior assumptions or weaker data-driven identification. For instance, the posterior distributions of ϕ_{earnings} (effect of earnings on consumption) and θ_{dividend} (variance in dividend yields) remain relatively aligned with their priors, reflecting limited variance in the corresponding observables or shorter sample periods. These findings underscore the importance of carefully choosing priors that are both theoretically justified and empirically plausible, especially for parameters associated with less informative data. Policymakers relying on these parameters must exercise caution and consider integrating additional indicators to strengthen the robustness of policy recommendations.

The robustness analysis further reveals that many parameters demonstrate strong resilience to prior variations, ensuring reliable policy design. For example, sensitivity analysis shows that reducing the prior mean of γ from 0.5 to 0.3 results in only minor changes to its posterior median, suggesting strong data-driven identification. Similarly, posterior predictive checks confirm the model's ability to reproduce key data moments across alternative prior specifications, particularly for parameters like ρ_{vix} , γ , and ϕ_{housing} . These parameters are strongly shaped by the empirical properties of financial and macroeconomic data, highlighting the effectiveness of the extended dataset in enhancing parameter identification. However, the results also highlight the need for dynamic policy frameworks that account for structural changes and parameter uncertainty. For example, while ϕ_{housing} robustly estimates the housing wealth effect, its relevance may evolve due to demographic shifts or changes in financial markets. Similarly, the persistence of financial volatility, as captured by ρ_{vix} , could vary with regulatory reforms or market developments, necessitating continuous reevaluation.

From a policy perspective, the findings underscore the interconnectedness of financial markets and household consumption in driving macroeconomic outcomes. The robust identification of key parameters, particularly those associated with uncertainty and credit shocks, emphasizes the need for proactive policies to stabilize financial conditions. Countercyclical measures such as liquidity provision, targeted credit facilities, and forward guidance can mitigate the adverse effects of financial shocks on consumption and aggregate demand. Furthermore, stabilizing housing markets through mortgage refinancing programs or property tax adjustments can amplify these effects by supporting household wealth and spending.

In conclusion, the extended DSGE model provides a comprehensive framework for understanding the macroeconomic implications of financial and consumption dynamics. By incorporating new observables and carefully analyzing parameter robustness, the model offers insights that enhance the reliability of policy interventions. This balance between prior-driven and data-driven identification strengthens the credibility of the model's findings and ensures that policy recommendations are both theoretically sound and empirically robust.

5 Conclusion

This study extends the standard DSGE framework by incorporating a broader set of observables, including VIX, earnings, savings, credit, housing, durables, and dividends, to better capture the dynamic interplay between financial markets and macroeconomic behavior. By expanding the model's informational content, the inclusion of these observables enhances the identification of key structural parameters, providing richer insights into the transmission mechanisms through which financial shocks propagate to the real economy. The estimation results demonstrate the persistent nature of financial market shocks, the sensitivity of household consumption to uncertainty and credit availability, and the significant wealth effects associated with housing. These findings highlight the interconnectedness of financial markets and macroeconomic stability, offering practical implications for designing effective monetary and fiscal policy interventions.

Key insights include the robust identification of persistence parameters for financial shocks, such as ρ_{vix} , ρ_{credit} , and ρ_{housing} , which reflect the prolonged influence of volatility, credit, and housing market conditions on macroeconomic outcomes. Sensitivity parameters, including γ (the effect of uncertainty on consumption) and ψ (the effect of credit conditions on consumption), reveal the critical role of financial stability in maintaining aggregate demand. The strong empirical identification of these parameters underscores the importance of countercyclical policies, such as forward guidance, liquidity provision, and targeted credit market interventions, in mitigating the adverse effects of financial disruptions.

The estimated effect of housing wealth, captured by ϕ_{housing} , confirms the significant influence of housing markets on household consumption, reinforcing the need for policies that stabilize housing wealth during periods of market volatility. Additionally, while earnings (ϕ_{earnings}) exhibit a weaker direct impact on consumption, this parameter highlights the potential benefits of addressing broader income distribution issues to support household spending. However, parameters associated with less informative observables, such as θ_{dividend} , exhibit greater sensitivity to prior assumptions, reflecting the importance of careful prior selection and the potential need for alternative indicators to strengthen identification.

The robustness analysis demonstrates that the extended DSGE model balances prior-driven and data-driven estimation effectively. Posterior predictive checks confirm the model’s ability to reproduce observed data moments across various prior specifications, particularly for parameters associated with financial shocks and consumption dynamics. However, sensitivity to priors for parameters tied to observables with low variance or limited sample periods highlights the need for caution when relying on these estimates for policy design. Policymakers must acknowledge these uncertainties and consider integrating additional data or adopting flexible, dynamic policy frameworks to address evolving economic conditions. The policy implications derived from this analysis emphasize the critical role of stabilizing financial markets, credit conditions, and housing wealth in supporting consumption and mitigating macroeconomic volatility. Targeted interventions, such as mortgage refinancing programs, property tax adjustments, and enhanced credit access, can amplify the stabilizing effects of broader monetary and fiscal policies. Furthermore, addressing uncertainty through central bank transparency and forward guidance can reduce consumption declines during periods of heightened financial market stress.

Building on the findings of this research, there are several promising avenues for future work. First, expanding the set of observables to include global financial variables, such as cross-border capital flows or international trade shocks, could offer a more comprehensive understanding of the global financial transmission mechanisms. This extension would enable policymakers to evaluate the interconnected effects of domestic and international financial developments. Second, incorporating nonlinear dynamics or regime-switching frameworks into the DSGE model could provide a more nuanced understanding of the asymmetric effects of shocks during crises and recoveries. For example, the impact of uncertainty on consumption may differ during periods of financial stability versus periods of heightened market stress. Nonlinear modeling approaches could capture these dynamics more accurately, offering deeper insights into crisis management.

Third, exploring heterogeneity across households, regions, or industries could yield more targeted policy insights. For instance, the differential impact of financial shocks on low- versus high-income households or on credit-constrained versus credit-unconstrained firms could inform the design of more equitable and effective policy measures. Such heterogeneity could also improve the model’s ability to capture the distributional effects of shocks and policies. Finally, integrating high-frequency data, such as daily financial market indicators or real-time survey data, could enhance the timeliness and accuracy of the model’s forecasts. High-frequency data would allow policymakers to monitor evolving economic conditions more effectively and respond to emerging risks with greater precision. This could be particularly valuable in addressing sudden and severe shocks, such as those experienced during global financial crises or pandemics.

In summary, while this study provides a robust framework for analyzing the macroeconomic implications of financial and consumption dynamics, future work could further enrich the model’s applicability and relevance. By incorporating additional observables, exploring nonlinearities and heterogeneities, and leveraging high-frequency data, future research can enhance the capacity of DSGE models to inform policy design in an increasingly complex and interconnected economic environment.

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Appendix

A1. Prior and Posterior Density

The results presented in Figure 1 illustrate the posterior distributions of the estimated parameters alongside their prior means. The posterior densities, represented by solid black lines, generally exhibit a clear shift and tightening relative to the prior means (green dashed lines), indicating that the data strongly informs the parameter estimates. For instance, parameters like α (capital share) and β (household discount factor) show well-defined and sharp posterior distributions, reflecting strong identification from the data. Similarly, persistence parameters such as ρ_R , ρ_z , and ρ_{housing} reveal high persistence in shocks to monetary policy, technology, and housing markets, respectively.

Interestingly, the posterior distributions for parameters related to financial and consumption observables, such as ρ_{vix} , ρ_{earnings} , and ρ_{credit} , highlight the role of financial volatility, earnings, and credit growth in explaining macroeconomic fluctuations. While the posterior of ρ_{vix} suggests moderate persistence, the results for ρ_{savings} demonstrate extreme persistence, which is consistent with the smooth behavior of savings dynamics in the data. Overall, these findings underscore the critical role of financial and consumption-related shocks in influencing macroeconomic outcomes. The posterior estimates provide robust evidence for the interconnected dynamics of financial volatility, credit markets, and aggregate consumption, offering valuable insights for understanding the transmission of shocks and designing stabilization policies.

A2. Collinearity Analysis of Parameters

Figure 2 illustrates the degree of linear dependency among estimated parameters within the Bayesian framework of the DSGE model. Collinearity is assessed through prior means, highlighting potential identification issues that arise when parameters are interdependent.

The first plot represents collinearity patterns when a single parameter is considered in isolation. Dark red squares indicate high collinearity values close to 1, suggesting strong linear dependence between the respective parameters. For instance, parameters such as ρ_w and ρ_b , as well as ρ_g and ρ_μ , exhibit significant collinearity. This pattern suggests that these parameters may not be separately identified from the data without strong prior information, as they contribute similarly to the observed dynamics. Light orange and blue areas indicate relatively low collinearity, signaling better identification for these parameters.

The second plot extends the collinearity analysis to patterns involving two parameters. When two parameters are considered jointly, the heat map reflects the extent to which their combined contributions align with the model's dynamics. Here, we observe intensified collinearity patterns, particularly among parameters related to persistence (ρ_w, ρ_b, ρ_μ) and sensitivity ($s2prime, \xi$). For example, persistence parameters exhibit high joint collinearity values, shown by dark red squares, highlighting that the data struggles to differentiate between their effects. Parameters such as ρ_{savings} and $\rho_{\text{dividends}}$ show noticeable collinearity with others, likely stemming from overlapping dynamics of savings and dividends in influencing consumption and financial stability. The observed collinearity patterns underscore the importance of prior selection in addressing identification challenges. These results further emphasize the role of additional observables—such as VIX, housing wealth, and credit growth—in breaking parameter dependence and improving identification within the DSGE model.

A3. Identification Strength and Sensitivity Analysis of Structural Parameters

Figure 3 presents a detailed analysis of parameter identification strength and sensitivity within the Bayesian DSGE model, leveraging the asymptotic information matrix. The analysis evaluates two key components: (i) the *identification strength*, reflecting how well each parameter is identified relative to its prior and posterior uncertainty, and (ii) the *sensitivity component*, which quantifies the contribution of each parameter to the likelihood function relative to the prior information. Both components are presented on a log scale for interpretive clarity.

In the *top panel*, identification strength is depicted for all estimated parameters. Bars indicate the relative strength of identification, with blue bars showing values relative to the posterior parameter value and red bars relative to the prior standard deviation. Parameters such as ρ_{savings} , ρ_{durables} , and ρ_{earnings} exhibit the highest identification strength, highlighting that these parameters are well-identified by the data, with minimal reliance on prior information. Conversely, parameters such as $\rho_{\text{dividends}}$, θ_μ , and ϕ_x show relatively weaker identification, suggesting that these parameters require informative priors to compensate for limited data-driven information.

The *bottom panel* displays the sensitivity component of the asymptotic information matrix, reflecting how the log-likelihood function responds to changes in parameter values. Parameters such as ρ_{savings} , ρ_{durables} , and ρ_{housing} exhibit high sensitivity, highlighting their strong contribution to model dynamics. Conversely, parameters like α , β , and ρ_w exhibit negative sensitivity, emphasizing the need for careful prior specification to prevent distortions in their posterior distributions.

These findings reinforce the importance of incorporating additional observables and leveraging informative priors to improve parameter identification. Researchers and policymakers can use these insights to prioritize the estimation of well-identified parameters and exercise caution with those requiring stronger prior assumptions.

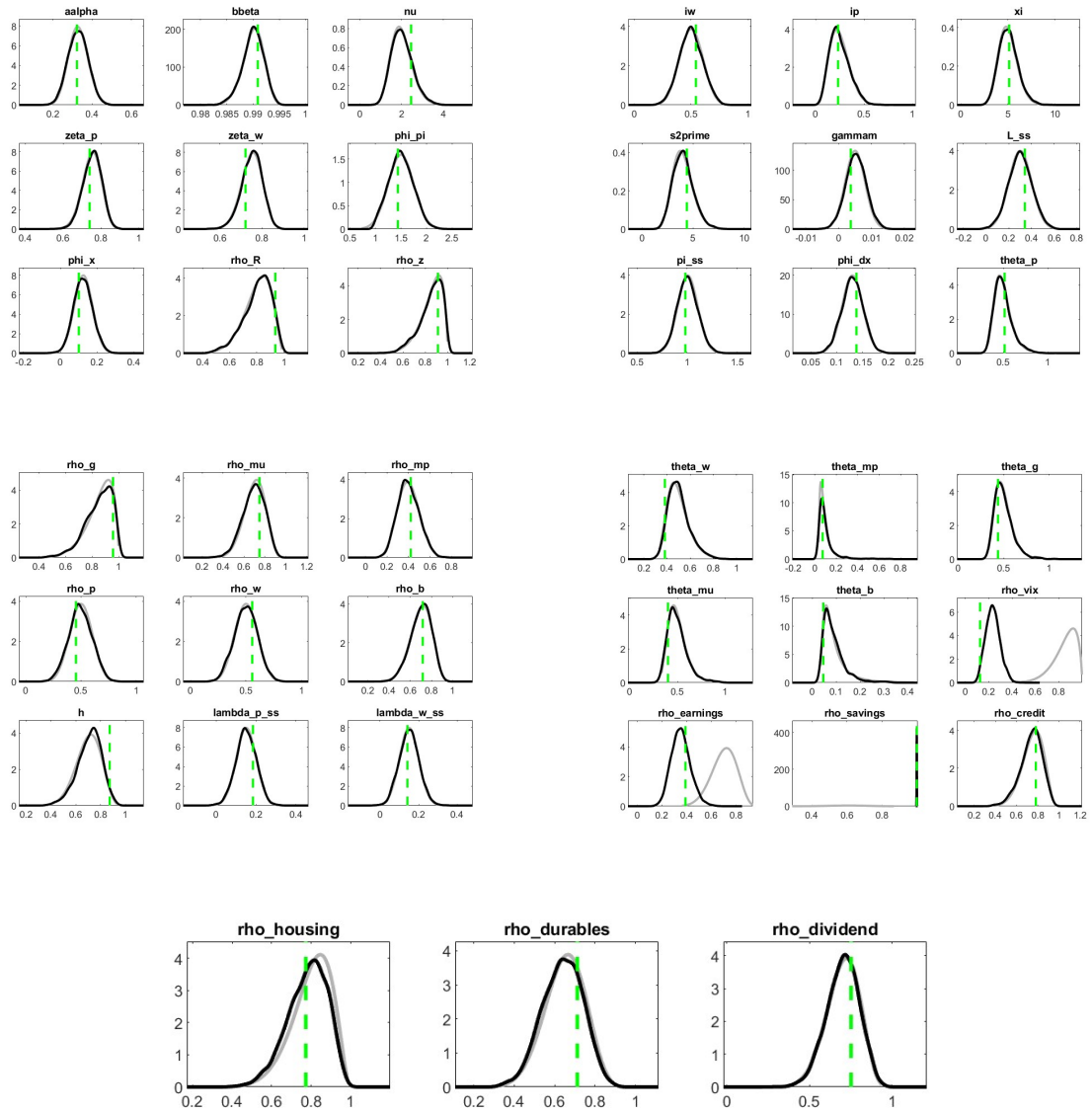
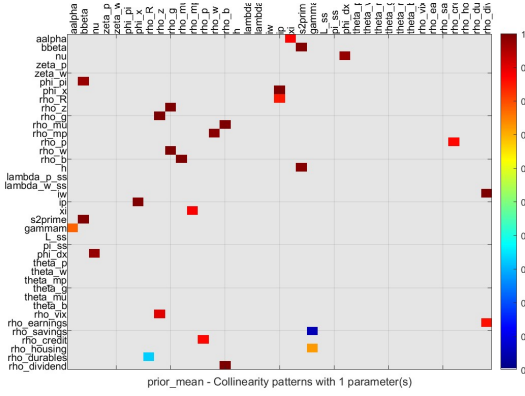
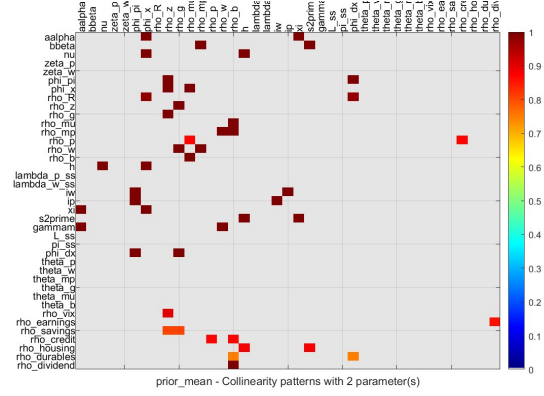


Figure 1: Posterior distributions for estimated parameters.



Collinearity patterns with a single parameter.



Collinearity patterns with two parameters.

Figure 2: Collinearity Analysis of Estimated Parameters.

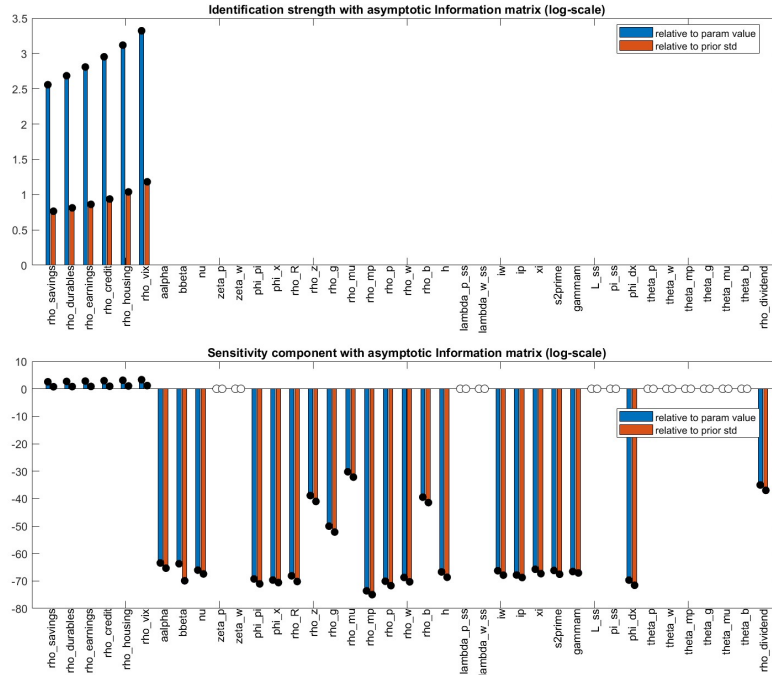


Figure 3: Identification strength (top) and sensitivity component (bottom) using the asymptotic information matrix. Blue bars indicate values relative to parameter estimates, and red bars reflect values relative to prior standard deviations.